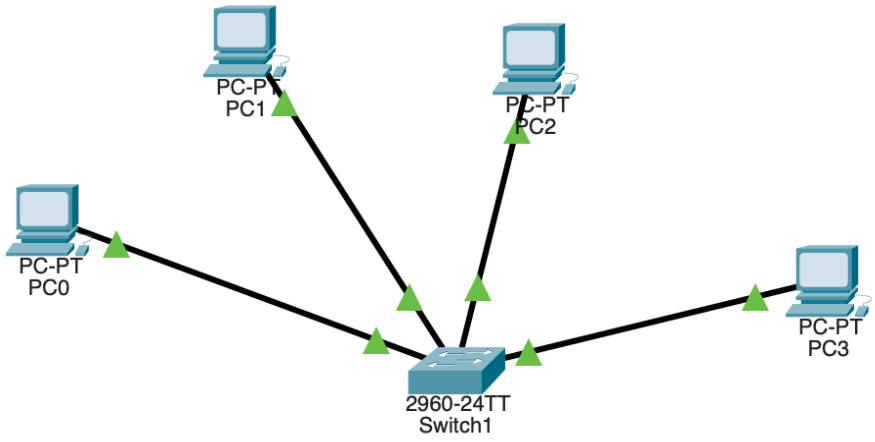
Task 02

1. Multiple PCs in the Same Network



IP Addressing

Device	IP Address	Subnet Mask
PC-0	192.168.10.10	255.255.255.0
PC-1	192.168.10.20	255.255.255.0
PC-2	192.168.10.30	255.255.255.0
PC-3	192.168.10.40	255.255.255.0

Tasks

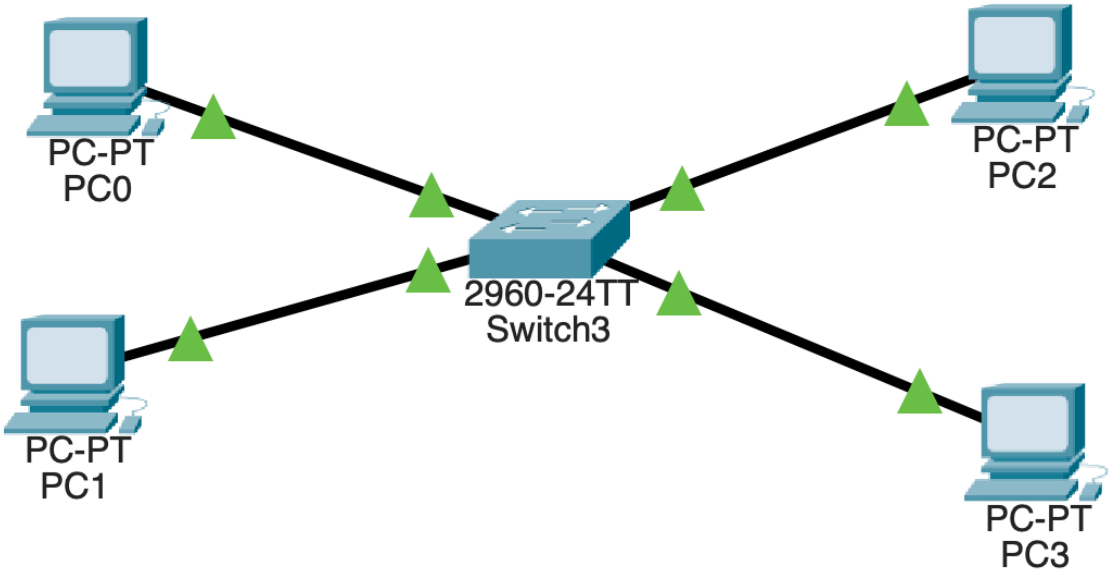
1. Add **4 PCs** and **1 switch**.
2. Connect all PCs to the switch.
3. Configure the IP addresses.
4. Configure the subnet mask.
5. From PC-0, ping PC-1.
6. From PC-0, ping PC-2.
7. From PC-0, ping PC-3.
8. Record the results.

2. Same Switch, Different Networks

A switch connects computers belonging to **two different departments**:

- Department A → Network 192.168.10.0/24
- Department B → Network 192.168.20.0/24

There is **no router** in this topology.



IP Addressing

Device	IP Address	Subnet Mask	Network
PC-0	192.168.10.10	255.255.255.0	192.168.10.0/24
PC-1	192.168.10.20	255.255.255.0	192.168.10.0/24
PC-2	192.168.20.10	255.255.255.0	192.168.20.0/24
PC-3	192.168.20.20	255.255.255.0	192.168.20.0/24

Tasks

1. Build the topology.
2. Configure all four PCs.
3. Determine the network of each PC.
4. Predict which PCs should communicate.
5. Use ping to test your predictions.
6. Take a screenshot of one ping request to the same network and one to a different network.

3. Connecting Two Different Networks Using a Router

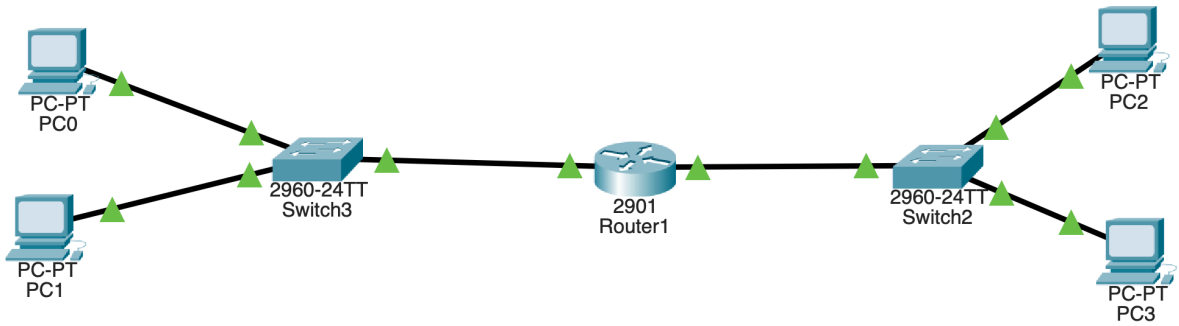
In this task, you will connect two different IP networks using a router. You will configure IPv4 addresses on PCs and router interfaces, assign default gateways, and test communication between the networks.

Consider a university with two computer labs.

- **Lab A** uses the network 192.168.10.0/24.
- **Lab B** uses the network 192.168.20.0/24.
- A router is required to allow communication between the two networks.

The topology will contain:

- 4 PCs
- 2 switches
- 1 router



IP Addressing

We will use two separate /24 networks.

Network A: 192.168.10.0/24

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.20	255.255.255.0	192.168.10.1
Router G0/0	192.168.10.1	255.255.255.0	—

Network B: 192.168.20.0/24

Device	IP Address	Subnet Mask	Default Gateway
PC2	192.168.20.10	255.255.255.0	192.168.20.1
PC3	192.168.20.20	255.255.255.0	192.168.20.1
Router G0/1	192.168.20.1	255.255.255.0	—

- Configure Router
- Test Communication Within Network A
- Test Communication Within Network B
- Test Communication between Network A and Network B.